Possibilistic Decomposition of a Tomographic Inversion

Separating data-forced structure from regularization artifact

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What this is

A tomographic image is not a picture of the Earth. It is the output of a long chain of lossy, selective steps—sparse rays, a simplified forward model, a parameterization, a regularizer, an optimizer—and what survives that chain is not the structure but a *filtered residue* of it. You and I have said as much to each other already. The practical question that framing forces is the one this note is about:

> ***Given a finished inversion, which features did the data force, and which did the regularization invent?***

Standard practice does not answer that question. It picks a damping value, returns one model, and attaches a posterior covariance—and every part of that is conditional on the damping choice. Your own thesis says it plainly: "there is no simple solution for regularization, and optimization of damping conditions remains a highly parametrization and input dependent problem" (ZTM/TFM, §11).

This note proposes a different reading of an inversion—a **possibilistic** one—and demonstrates it end-to-end on synthetic travel-time tomography with two forward operators, straight-ray and Eikonal. It is a methods communication and a draft; the demonstrations are synthetic; nothing here is claimed as proven. What I am claiming is that the reading is sound, that it is implemented and validated against known ground truth, and that the place it stops working cleanly is a real open problem worth our working on together.

1 The problem: a damping choice is not a fact

Tomographic inversion is ill-posed. Many models fit the data within noise. To return *one* model you must regularize—damp, smooth, prefer a reference—and the model you get is as much a property of that choice as of the data.

The honest consequence: a feature in a tomographic image belongs to one of three classes, and they are not the same kind of thing.

A posterior covariance does not draw this line. It reports a spread *under one prior and one*

Class	Meaning
Forced	Present in *every* model consistent with the data and the hard physical bounds. Data-determined. Independent of the damping choice.
Forbidden	Present in *no* such model.
Measure-dependent	Present in *some* consistent models and absent in others. The data do not determine it—which sign or structure you see is set by the regularization. Such a feature may still be physically real: measure-dependence diagnoses *underdetermination*, not falsehood.

damping; it cannot tell you that a given feature would vanish under a different, equally defensible choice. The damping problem of §11 is, in this language, a symptom of reading an inversion through the wrong layer: it is the search for the "right" measure-layer choice to extract a model, when the features that *depend* on that choice are exactly the ones the data does not determine.

2 The possibilistic frame

The distinction in §1 is the **possibilistic / probabilistic** split, taken from the two-layer discipline of the Closure Forces Structure programme (`inverse_born_methodology.md`) and carried over to inversion:

- The **possibilistic layer** asks what is *forced or forbidden* by the data and the hard physical constraints alone. Its answers are unconditional—they do not depend on a prior, a damping value, or a measure.
- The **probabilistic layer** asks how a measure distributes weight over what the possibilistic layer permits. Its answers are conditional on that measure.

A posterior covariance is a probabilistic-layer object. The possibilistic layer is the one that answers the §1 question, and it is the one standard tomography leaves on the table.

One clarification carries the rest. Throughout, F is the *hard* admissibility set—every model that fits within the noise tolerance and obeys the hard physical bounds—and the probabilistic layer is taken as conditional on that admissibility. With a smooth likelihood a posterior need not vanish sharply outside F; the two-layer reading treats F as the tolerance-defined region and a measure μ as a measure on it. Every "measure on F" below—and every bound stated in terms of "F's extent," including Figure 3's—is a claim about admissible μ in that conditional sense, not about an untruncated smooth-likelihood posterior.

The split has a natural geometry, and the next three figures build it up one idea at a time.

Figure 1 — the feasible set. The data and the hard constraints do not pick out a model; they pick out a *set* F of models—every model that fits within noise and respects the bounds. F is a set, not a probability density—the figures shade it only to mark the region. Project F onto a feature's axis and the feature falls into one of two classes. It is **forced** when the projection lies entirely on one side of zero: every model consistent with the data agrees on its sign, and no prior is needed to settle it. It is **measure-dependent** when the projection straddles zero: the data

leave the sign open, and any single value reported there is fixed by the measure you add, not by the data.

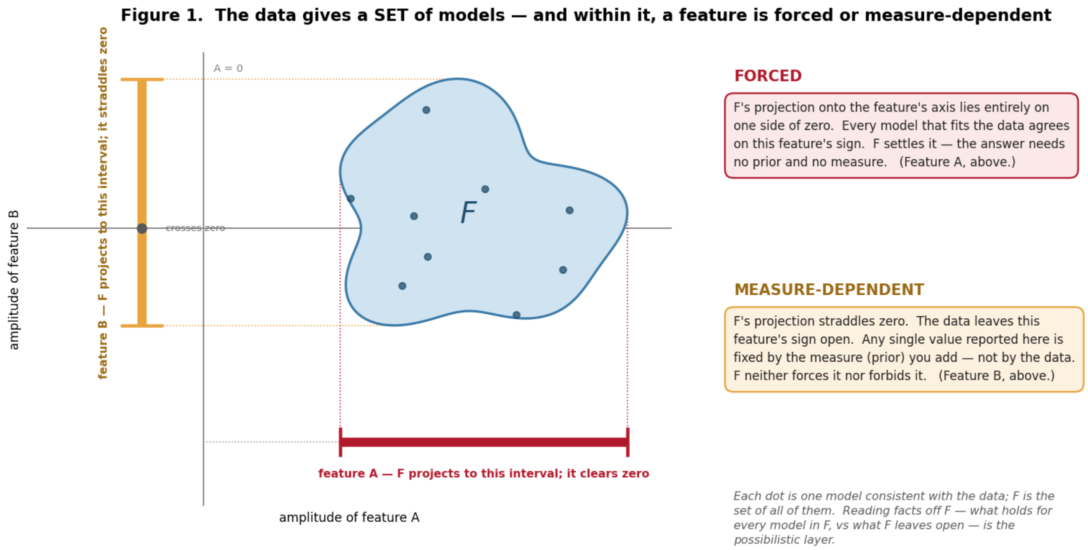

Figure 1: The data give a set, not a point. A feature is forced when F's projection clears zero and measure-dependent when it straddles zero—the two terms this note turns on, defined geometrically.

On the word "possibilistic." I use it in a deliberately narrow sense—the crisp-set limit of possibility theory, equivalently a set-valued feasibility reading, not a graded possibility distribution. F is an ordinary feasible set: a model is in it or it is not. A feature is *forced* exactly when a fact holds for every model in F (necessity 1, in possibility-theory terms) and *measure-dependent* when F neither forces nor forbids it. No fuzzy membership is invoked; the only structure used is the set F and the all-or-nothing quantifier over it.

Figure 2 — two reports. The same F can be turned into a reported answer two ways. The *possibilistic report* uses exactly F: forced features pinned to a sign, measure-dependent features returned as intervals—no more information than F carries, and no less. The *Bayesian report* uses F plus a measure μ—it places μ on F and reports its mean and credible interval. μ is information beyond F, and two defensible choices of μ can disagree on a measure-dependent feature's sign, so that choice needs a justification of its own. On a forced feature the two reports agree; the divergence is confined to the measure-dependent ones.

Figure 3 — the layers compose. The two are not rivals run side by side; they compose in sequence. The possibilistic layer runs first and bounds which measures are admissible—a measure is admissible only if it is supported on F. The Bayesian layer then runs inside that bound: across all admissible measures, the *attainable range of reported answers*—the measure-uncertainty—cannot exceed F's extent. On a forced feature that bound clears zero, so no admissible measure can move the sign; on a measure-dependent feature it straddles zero and the sign is genuinely open. Possibilism does not compete with Bayes—it brackets it.

The object that carries the possibilistic content is the **feasible interval**. Take an ensemble of models that each fit the data within noise and respect the hard bounds. For each cell, record the

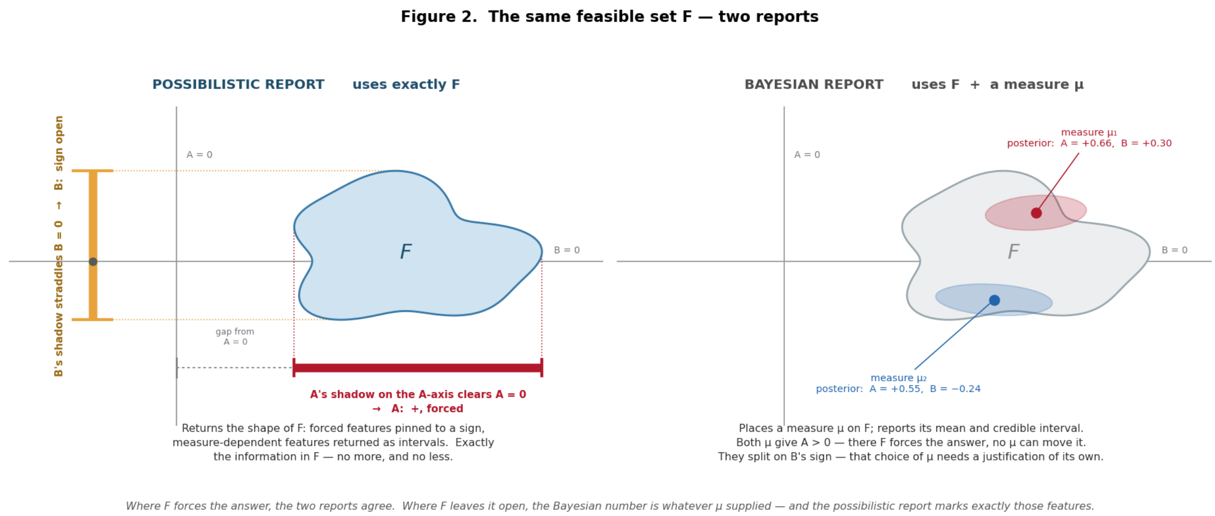

Figure 2: Two reports from one feasible set. Possibilistic: use exactly F. Bayesian: use F plus a measure μ—defensible μ agree where F forces the answer and can conflict where it does not.

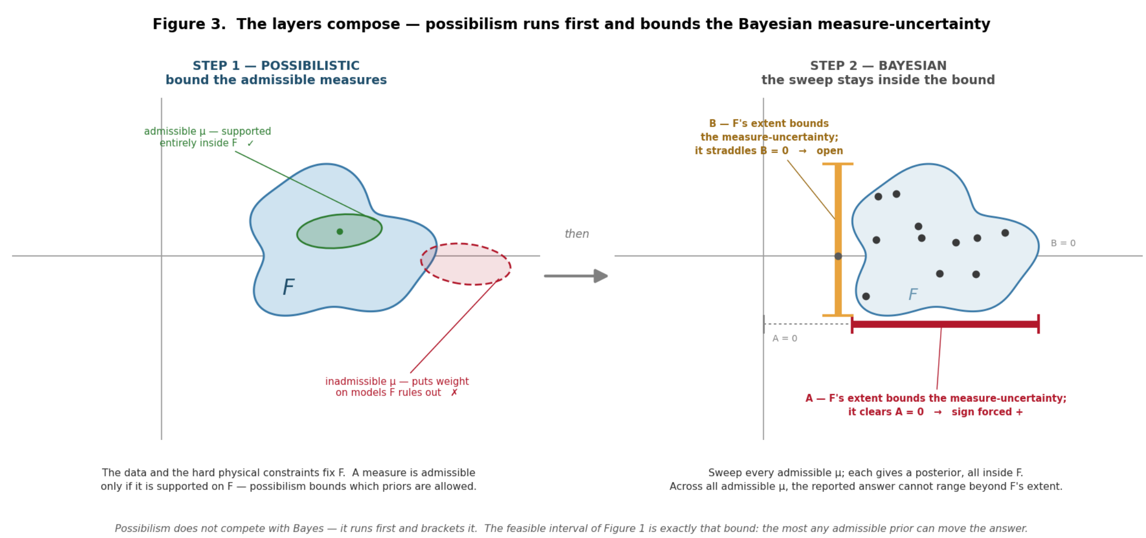

Figure 3: The layers compose. Possibilism runs first and bounds the admissible measures; the Bayesian sweep then stays inside that bound. The feasible interval below is exactly that bound.

interval $[a_{\min}, a_{\max}]$ of the anomaly across the ensemble. That interval is F projected onto the cell's axis—exactly the bound Figure 3 draws:

- $a_{\min} > 0$ everywhere in the ensemble $\rightarrow$ **forced-high** (no data-consistent model makes this cell non-positive);
- $a_{\max} < 0 \rightarrow$ **forced-low**;
- the interval straddles zero $\rightarrow$ **measure-dependent** (the sign itself is not data-forced);
- the interval sits inside a small band around zero $\rightarrow$ **forced-quiet**.

Figure 4 puts the classification on a concrete ensemble. The same ensemble, read two ways: the probabilistic reading collapses it to a mean and an error band; the possibilistic reading keeps the feasible interval and classifies it. The big feature is forced; the flanks and the small bump are measure-dependent—and the probabilistic band does not distinguish them.

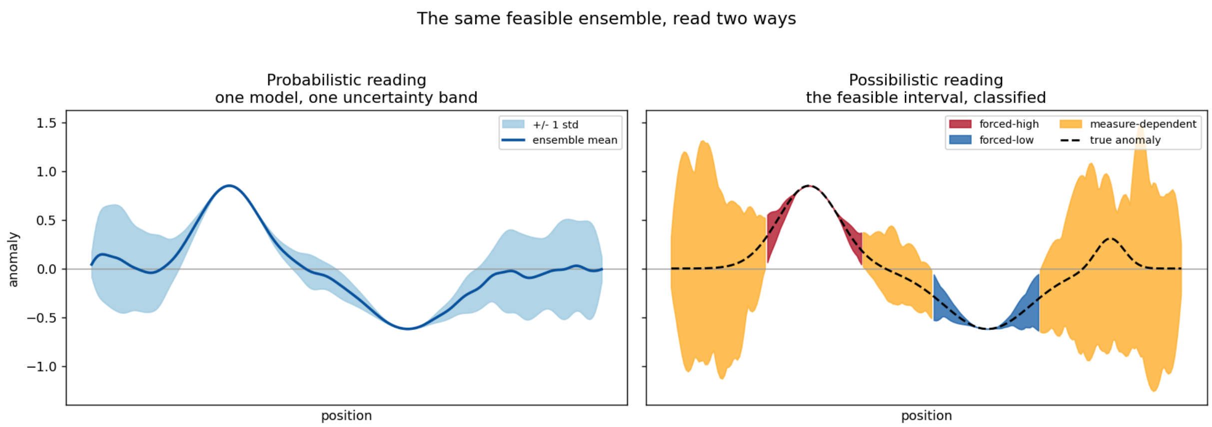

Figure 4: The same feasible ensemble, read two ways. Left: one model, one uncertainty band. Right: the feasible interval, classified into forced-high / forced-low / measure-dependent.

On prior art — said straight. This is not unprecedented, and pretending otherwise would not survive five minutes of your scrutiny. Computing bounds on what a model *can* be, rather than a single estimate, is the spirit of Backus–Gilbert extremal inversion (1968); the under-determined directions are the null space of resolution analysis; multi-model joint coupling has the Gramian-constraint literature (Zhdanov, 2012 onward). And a transdimensional / reversible-jump MCMC posterior ensemble (Bodin & Sambridge, 2009) already *contains* this information—forced structure appears as high-consensus, measure-dependent structure as sign-variable posterior mass. The epistemic intuition is old. What I am putting forward is narrower: not a new inverse-theory principle but a *reporting discipline*—the forced / measure-dependent split made an explicit, first-class output; measure-dependence treated as the operational diagnostic of *underdetermination*; and the whole run as a transferable, documented procedure on top of arbitrary inversion machinery, rather than the informal robustness check practitioners already perform inconsistently. The contribution is the operationalization, not the distinction.

3 Method

The feasible set. A model is *feasible* if it (a) fits the data to within the noise level and (b) satisfies the hard physical bounds. The bounds are not left implicit: `geophysical_invariants.md` stratifies them into Tier-1 (frame-independent—positivity, first-arrival minimality, a generous velocity envelope; used freely), Tier-2 (frame-dependent—typical mantle/crust values, reference Earth models, petrophysical relations; soft priors only), and Tier-3 (the observational context). The possibilistic feasible set is bounded by Tier-1 alone. Worth noting in passing: TFM currently clamps node velocities to 7.5–9.5 km/s and initializes from IASP91—Tier-2 values used as hard limits. That is the layer-confusion the stratification is meant to catch.

Sampling it. Generate many feasible models by inverting toward many random reference models, each inversion taken to the point where its misfit equals the noise level. The references diversify the data-null directions; the data-resolved directions are common to all. Then read the feasible interval and classify it (§2).

The decomposition is forward-model-agnostic. It operates on the ensemble of models; it does not care how they were produced. That is the point of the two demonstrations below: the *same* decomposition code runs on a linear straight-ray operator and a nonlinear Eikonal operator.

4 Demonstration 1 — straight-ray operator (linear)

A synthetic velocity model: one large tilted high-velocity slab, one broad low-velocity zone, three small sub-resolution blobs. Synthetic first-arrival times with 1.2% noise, dense four-edge ray coverage. The feasible set is sampled exactly per §3 (the linear operator admits an exact eigendecomposition, so the feasible set is parametrized cleanly).

Figure 5 is the result. The forced-sign cores sit on the true features—panel (e), the black forced-high contour lies inside the true slab, the white forced-low contour inside the true low-velocity zone. The numbers:

- **Forced cores are correct.** $\sim$7 sign errors within the resolution length out of $\sim$300 forced cells ($\sim$2%). What the method certifies, the ground truth bears out.
- **The measure-dependent shell captures $\sim$89% of the sub-resolution blob cells**—it correctly flags the detail the data cannot pin down.
- **The forced set is conservative**: $\sim$19% of the grid is sign-certain. This is not a weakness. It is the method saying out loud what is true—most of a tomographic image is *not* forced.

5 Demonstration 2 — Eikonal operator (nonlinear)

The straight-ray operator is exact only in a homogeneous medium. The faithful operator is the Eikonal first-arrival solver—the operator class your `FMM.cpp` implements. First-arrival rays bend: toward fast structure, away from slow (Figure 6). `eikonal.py` provides it (Fast Marching Method solver + ray-path Fréchet kernel), standalone and self-tested.

Because travel time is now a nonlinear functional of slowness, the one-shot linear solve is replaced by an iterative Levenberg–Marquardt inversion that recomputes the ray paths through the cur-

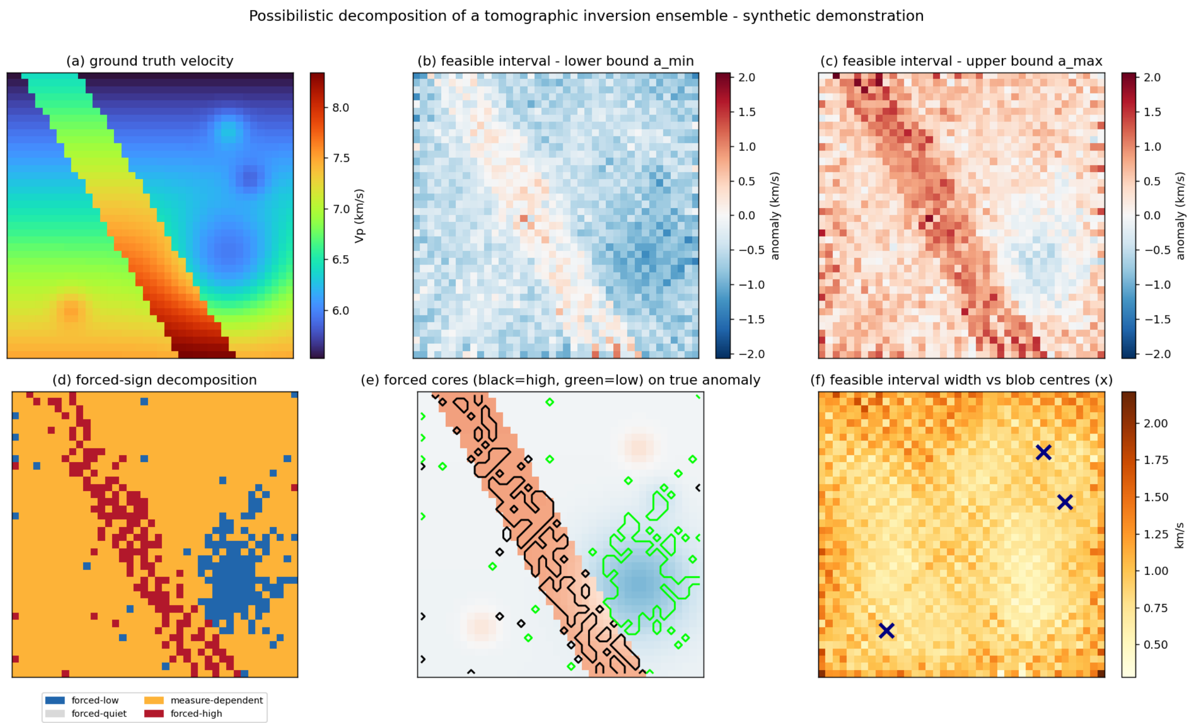

Figure 5: The straight-ray (linear) demonstration. Panel (d) is the decomposition; panel (e) shows the forced cores landing on the true features.

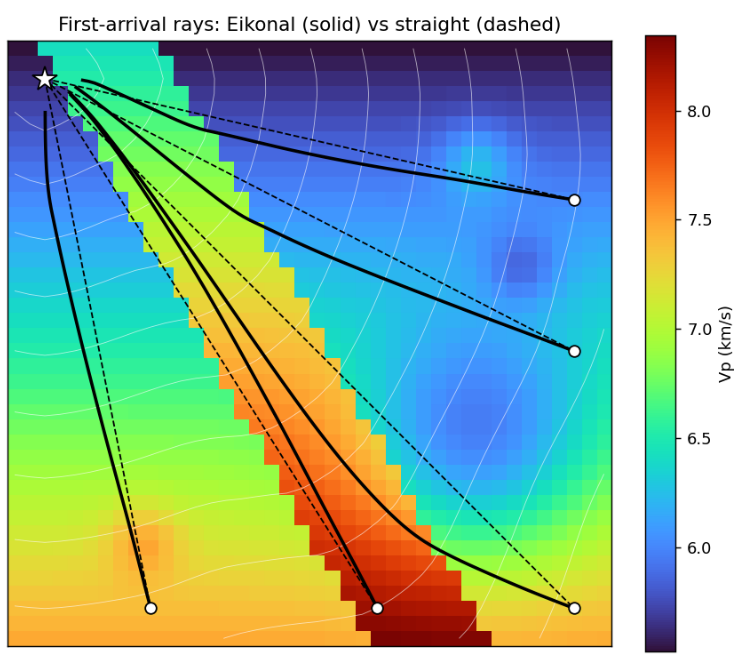

Figure 6: Why the forward operator matters. Solid: Eikonal first-arrival rays, bending through the medium. Dashed: the straight-ray approximation.

rent model—the DGN + FMM structure of your own pipeline. Figure 7 is the result, and the decomposition code is *identical* to Demonstration 1:

- **Forced cores ∼89% sign-correct within the resolution length**, but ∼71–81% strict (cell-exact). The gap is resolution-length blur—a forced cell one or two cells off a true feature edge, not a sign mistake—but the strict figure is the honest co-headline; the within-resolution number alone overstates the precision.
- **The measure-dependent shell captures ∼76% of the blobs.**

These figures are from the feasible-set sampler at its default operating point, and at that operating point the forced set itself over-claims: an adversarial stress test (§6, point 6; `witness_pass.md`, RWC-2) finds ∼41% of forced cells admit a feasible, equally-smooth, opposite-sign model. The decomposition layer is sound and forward-model-agnostic; what is coverage-limited is the feasible-set *sampling* that feeds it (§7).

The decomposition transferred across two genuinely different forward operators without change. That is the load-bearing result of this note: the possibilistic reading is a property of *inversions*, not of a particular operator.

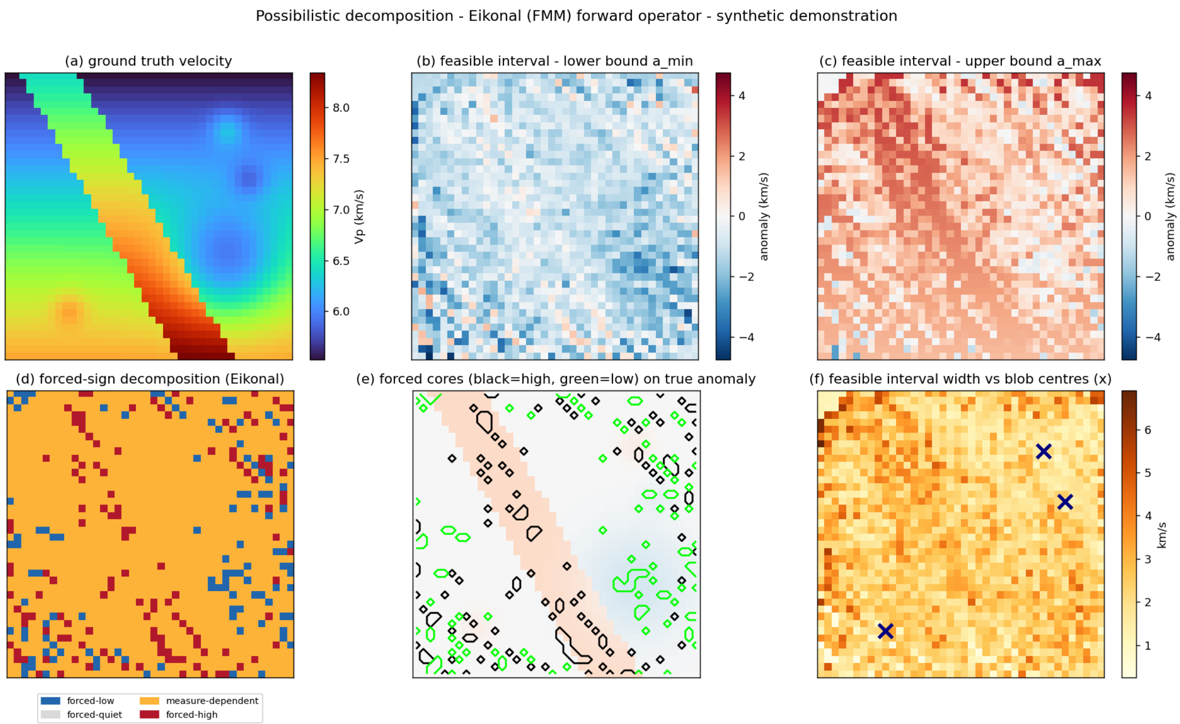

Figure 7: The Eikonal (nonlinear) demonstration—same decomposition, faithful operator.

6 What building it surfaced — the discipline the method needs

A method that always says "yes" is a gimmick. This one has real failure modes, and finding them is how I know it has content. Each was surfaced by a synthetic run failing honestly; each is now documented in the script headers.

1. **The ensemble must sample the feasible set's diversity, or it certifies its own bias.** An ensemble built by varying damping *strength* alone shares a bias in the data-null directions; the intersection then stamps that shared bias "forced." The fix is many random references.
2. **Sample where the data is actually fit**—at the misfit = noise level. An over-damped ensemble has had data-resolved structure regularized away.
3. **"Forced" is a sign statement off the interval, not a hard threshold.** A single anomaly-magnitude cutoff is brittle; the feasible interval is the honest object.
4. **Precision is honest only at the resolution length.** Every feasible model inherits the same forward-operator blur; the forced core is correct to within roughly one cell, not cell-exact.
5. **Feasible models must be physically plausible — smooth.** The raw null space of a ray (line-integral) operator is dominated by high-frequency checkerboard modes the operator cannot see. Those fit the data but are not admissible Earth models; perturbing freely in them injects unphysical speckle. A feasible model is data-fitting *and* smooth.
6. **The forced set is coverage-gated — and the gate is measurable.** A forced label asserts that *every* feasible model agrees; an ensemble that under-samples the feasible set will agree spuriously and over-claim. Two checks quantify this (`witness_pass.md`, RWC-1 and RWC-2): the forced set converges only above an ensemble-coverage threshold—here $N \approx 250$ sampled models for a 40×40 grid—and at the default operating point an adversarial stress test finds ∼41% of forced cells *false-forced* (a feasible, equally-smooth, opposite-sign model exists). A forced claim is honest only with its coverage-adequacy curve attached.

These are not patches. They are the methodology's anti-anchoring discipline—the same one that, in the Closure programme, keeps a derivation from quietly tuning itself to the answer it wants.

7 The open frontier — and where you come in

Here is the honest seam, and it is the reason this is a note to *you* and not a finished claim.

The linear straight-ray case has a clean, exact answer: the operator is fixed, $G^\top G$ eigendecomposes once, and the feasible set is parametrized directly. The nonlinear case does not. Sampling the feasible set of a nonlinear inverse problem—getting genuine diversity in the data-null directions without injecting artifacts—is itself a hard problem. Building Demonstration 2 walked straight into it: a naive Levenberg–Marquardt ensemble shares an inversion-dynamics bias; fixing that needs explicit, physically-constrained feasible-set sampling. I have a working version, but I would not call the nonlinear feasible-set sampler solved.

This is exactly the territory of transdimensional / reversible-jump MCMC tomography (Bodin & Sambridge, 2009 onward)—methods built precisely to produce a genuine posterior ensemble. And it is where ZTM already lives: your `runs=20` top-level Monte Carlo *is* a feasible-set sampler. A coverage study on the synthetic problem (`witness_pass.md`, RWC-1) now puts a number on the question: the forced set stabilizes only near $N \approx 250$ sampled models, and a 20-run ensemble sits at roughly three times the converged forced-set size—20 runs is far short of coverage adequacy. So the open question is no longer *whether* the sampler needs upgrading but *to what*: whether a transdimensional sampler reaches the coverage the forced/measure-dependent split requires, on real data, at tractable cost.

That question is not mine to answer from the outside. It needs your inversion expertise, your code, and your data. My contribution is the possibilistic reading and the formal scaffolding behind it; yours is the geophysics and the instrument. Neither half is sufficient alone. That is the collaboration I am proposing—not "adopt this method," but "here is a method, validated in the linear case, and here is the concrete nonlinear problem your own pipeline is already engaging. Let us solve that part together."

8 Honest scope

- Everything here is **synthetic**. The demonstrations validate the *method* against known ground truth; they are not a result about the real Earth.
- The Eikonal solver is **first-order FMM**. Its ∼2% mean accuracy is fine for a forward-model-agnostic demonstration; your FMM-VFD hybrid is the production-grade version.
- Two dimensions, modest velocity contrasts, a single noise realization. None of the conclusions depend on scale, but none have been *tested* at scale.
- The forced/measure-dependent split is **operator-relative**: it reports what is forced *given a forward operator*. An operator with systematic error propagates that error into the forced set. This is not a defect of the method—it is the method correctly reflecting the operator—but it means the operator's fidelity is load-bearing.
- The **forced set is coverage-gated**. It is trustworthy only above an ensemble-coverage threshold (RWC-1: $N \approx 250$ here); below it the sampler over-claims (RWC-2: ∼41% of forced cells false-forced at the default operating point). The demonstrations are reported at that default point and so over-state the forced set. The decomposition layer is sound—what is coverage-limited is the feasible-set *sampling* that feeds it; a forced claim must travel with its coverage-adequacy curve.

Pointers

- `geophysical_invariants.md` — the stratified constraint set.
- `synthetic_demo.py`, `synthetic_demo_eikonal.py` — the two demonstrations.
- `eikonal.py` — the standalone Eikonal forward operator (self-test: `uv run python eikonal.py`).
- `decomposition_exact.jl` — the decomposition layer formalized in exact `Rational{BigInt}` arithmetic (Julia); `classify` proven total, exclusive, and exhaustive, the feasible-interval properties exact-verified.
- `c/` — a dependency-free C port of the whole method: the forward operators, the Levenberg–Marquardt inversion over a hand-rolled dense-Cholesky module, the feasible-set sampler, and the decomposition, pulled together by `possibilistic_inversion.c`.
- `witness_pass.md` — the synthesis witness pass: an adversarial external review of the method by three independent models, and the two Required Witness Checks it carried, `rwc1_forced_stability.py` and `rwc2_disconnected_test.py`.
- `inverse_born_methodology.md` (Closure Forces Structure programme) — the source of the two-layer possibilistic / probabilistic discipline.
- Bodin, T. & Sambridge, M. (2009), *Seismic tomography with the reversible jump algorithm*, Geophys. J. Int. — the transdimensional-sampling reference.

- Abatchev, Z. (2019), ZTM/TFM, UCLA — the joint seismic+gravity inverter this note is in conversation with.